

Ashkan Shahbazi

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Nashville, TN - 37212, USA

SUMMARY

Machine Learning Researcher at the intersection of attention mechanisms, large language models, geometric deep learning, and applied mathematics. I develop and analyze mathematically grounded transformer and geometric architectures for large scale LLM/VLM/RL pipelines and neural simulators, including digital twin applications.

EDUCATION

- **Vanderbilt University** 2022 to 2027 (expected)
Ph.D. Candidate, Computer Science Nashville, TN, USA
 - Advisor: Prof. Soheil Kolouri
 - Research interests: Attention mechanisms, LLMs, geometric deep learning, digital twin simulators
- **Amirkabir University of Technology** 2016 to 2020
B.Sc., Biomedical Engineering Tehran, Iran
 - Thesis: ML based cuffless blood pressure estimation from bio signals

EXPERIENCE

- **Instacart** 🌱 Sep 2025 to Nov 2025
ML Ph.D. Intern, Content AI Team Remote (San Francisco, CA)
 - Designed and deployed a cheap→strong LLM cascade for catalog attribute extraction that reduced per item inference cost by roughly 75 to 80% (down to about 20 to 25% of an always o3 pipeline) while matching or slightly improving o3 level F1 across more than 50 attributes and around 150k labeled products.
 - Systematically benchmarked three confidence estimation families (self consistency, cross LLM agreement, verifier ensembles) on a 147k sample internal golden set.
 - Built a lightweight student verifier that predicts five level confidence scores through distillation from cross model and verifier ensembles; the student outperformed its ensemble teachers in F1 and cost aware cascade metrics while running at a fraction of their latency and compute.

PATENTS AND PUBLICATIONS [\[Google Scholar\]](#) C=CONFERENCE, P=PATENT, S=IN SUBMISSION

- [C.1] A. Shahbazi, E. Akbari, D. Salehi, X. Liu, N. Naderializadeh, S. Kolouri (2025). **ESPFormer: Doubly Stochastic Attention with Expected Sliced Transport Plans**. In *International Conference on Machine Learning (ICML)*.
- [C.2] A. Shahbazi*, K. Pereira*, J. S. Heiselman, E. Akbari, A. C. Benson, S. Seifi, X. Liu, G. L. Johnston, J. Y. Wu, N. Simaan, M. L. Miga, S. Kolouri (2026). **Neural Augmented Kelvinlet for Real Time Soft Tissue Deformation Modeling**. In *AAAI Conference on Artificial Intelligence (AAAI)*.
- [C.3] X. Liu, Y. Bai, R. D. Martín, K. Shi, A. Shahbazi, B. A. Landman, C. Chang, S. Kolouri (2025). **Linear Spherical Sliced Optimal Transport: A Fast Metric for Comparing Spherical Data**. In *International Conference on Learning Representations (ICLR)*, spotlight.
- [C.4] X. Liu, R. D. Martín, Y. Bai, A. Shahbazi, M. Thorpe, A. Aldroubi, S. Kolouri (2025). **Expected Sliced Transport Plans**. In *International Conference on Learning Representations (ICLR)*.
- [C.5] H. Tran*, Y. Bai*, A. Kothapalli*, A. Shahbazi, X. Liu, R. D. Martín, S. Kolouri (2024). **Stereographic Spherical Sliced Wasserstein Distances**. In *International Conference on Machine Learning (ICML)*, spotlight.
- [C.6] A. Kothapalli, A. Shahbazi, X. Liu, R. Sheng, S. Kolouri (2024). **Equivariant vs Invariant Layers: A Comparison of Backbone and Pooling for Point Cloud Classification**. In *ICML Workshop on Geometry grounded Representation Learning and Generative Modeling (GRaM)*.
- [S.1] A. Shahbazi*, P. He*, A. Abbasi, Y. Bai, X. Liu, E. Akbari, N. NaderiAlizadeh, S. Kolouri (2025). **LUNA: Linear Universal Neural Attention with Generalization Guarantees**. Manuscript under review.
- [S.2] A. Shahbazi, C. Thrash, Y. Bai, K. Hamm, N. NaderiAlizadeh, S. Kolouri (2025). **LOTFormer: Doubly Stochastic Linear Attention via Low Rank Optimal Transport**. *arXiv:2509.23436*.
- [S.3] H. Tran*, Y. Bai*, A. Shahbazi*, J. R. Hershey, S. Kolouri (2024). **Understanding Learning with Sliced Wasserstein Requires Rethinking Informative Slices**. *arXiv:2411.10651*.
- [S.4] A. Abbasi, A. Shahbazi, H. Pirsiavash, S. Kolouri (2025). **Vector Quantized Soft Label Compression for Dataset Distillation**. Manuscript under review.
- [S.5] A. Abbasi, A. Shahbazi, H. Pirsiavash, S. Kolouri (2024). **One Category One Prompt: Dataset Distillation Using Diffusion Models**. *arXiv:2403.07142*.
- [P.1] A. Shahbazi, K. Pereira, J. S. Heiselman, E. Akbari, A. C. Benson, S. Kolouri, M. I. Miga, N. Simaan (2025). **System and Method for Real Time Soft Tissue Simulation via Hybrid AI Driven Physics Informed Modeling**. U.S. patent pending, filed January 2025.

SKILLS

- **Programming Languages:** Python, C++, MATLAB, Java
- **Machine Learning Libraries:** PyTorch, JAX, scikit-learn, NumPy, SciPy, OpenCV, Pandas
- **Systems and GPU Computing:** CUDA, custom GPU kernel programming, PyTorch extensions
- **Cloud & Tools:** Linux, Git, Docker, AWS, GCP, Unity

HONORS AND AWARDS

- **Lambda Research Grant Program** Oct 2025
Lambda Cloud 
 - Awarded \$2,000 in cloud credits to support machine learning research on scalable attention mechanisms for large language and vision language models.
 - Selected through a competitive research grant program focused on high impact applied machine learning projects.
- **Ranked Top 0.1% in Iran’s National University Entrance Exam** Sep 2016
National University Entrance Examination, Iran
 - Ranked 335th among more than 140,000 students in the nationwide university entrance exam.

PROFESSIONAL SERVICE

- **Reviewer** 2025 to Present
AAAI 2026; NeurIPS 2025; ICLR 2025 to 2026
 - Reviewed submissions in machine learning, attention mechanisms, geometric deep learning, and large language models.
- **Mentor** Summer 2023
VISE Summer Fellowship Program, Vanderbilt University 
 - Mentored VISE fellow Zhanqi Zhu on research topics in machine learning and geometric modeling.
 - Guided project design, experiments, and presentation of results for a summer research project.

PROFESSIONAL MEMBERSHIPS

- **Vanderbilt Lab for Immersive AI Translation (VALIANT)**, AI Scholar 2025 to Present
- **Association for Computing Machinery (ACM)**, Member 2025 to Present
- **Institute of Electrical and Electronics Engineers (IEEE)**, Member 2025 to Present

TEACHING EXPERIENCE

- **Teaching Assistant, Geometric Deep Learning** Spring 2024
Vanderbilt University
 - Assisted with lectures, homework design, and project mentoring for a graduate course on geometric deep learning.
- **Teaching Assistant, Statistics and Probability** Spring 2021
Amirkabir University of Technology
 - Led problem solving sessions and supported exam preparation for an undergraduate course in statistics and probability.

TALKS AND PRESENTATIONS

- **Geometric Neural Simulators for Modeling Liver Tissue Deformation** Dec 2024
13th Annual Symposium on Surgery, Intervention, and Engineering, Vanderbilt University, Nashville, TN
 - Oral presentation on geometric neural simulators for organ level digital twin style deformation modeling.